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Summary

Postdoctoral researcher and research advisor specializing in robotics and artificial intelligence, with expertise in multi-agent pathfinding, deep reinforcement learning, and UAV trajectory planning. I recently held a postdoctoral position at MMMI, University of Southern Denmark, and earned my Ph.D. in Electrical Engineering from the University of Skikda. I developed HM-DRL, a method for enhancing multi-agent pathfinding with a heatmap-based heuristic for distributed deep reinforcement learning. I also developed Gen4jectory, a 4-D UAV trajectory planning algorithm that guarantees zero loss-of-separation. Also created FDA*, a focused grid-based pathfinding method that dramatically reduces memory use and run-time.

Skills

Languages: English, Arabic, French.
Coding: Python, PyTorch, TensorFlow, ROS, Gazebo, MATLAB, Git, LaTeX.
Misc.: Project planning.

Experience

Research Advisor - KongeDrones ApS (Oct 2025 - Present)
Herning, Denmark
Highlights:

- Advising on R&D directions for autonomous drone systems, with a focus on navigation, trajectory planning, and multi-UAV coordination.
- Supporting the translation of robotics and AI research into practical solutions aligned with product and business needs.

Postdoctoral Researcher - MMMI - University of Southern Denmark (2023-2024)
Odense, Denmark
Summary: As a postdoctoral researcher at SDU, I focused on developing multi-robot coordination approaches using deep reinforcement learning.
Highlights:

- Contributed to the "Swarm Robotics for Industry 4.0" project supported by the Independent Research Fund in Denmark, under grant 0136-00251B.

- Developed HM-DRL, a deep reinforcement learning approach for distributed multi-robot pathfinding (MAPF). Then I wrote and published a journal paper in Springer's Applied Intelligence journal.
- Developed Relocation-MAPF, a multi-robot pathfinding (MAPF) algorithm that enhanced both the efficiency and success rate of robot fleets using a relocation strategy that aims to reduce the number of bottlenecks in densely populated MAPF environments.

Robotics Researcher, Ph.D. Internship - University of Southern Denmark (2023-02)
Odense, Denmark
Highlights:

- Studied the application of deep reinforcement learning for swarms of AMRs in modern Industry 4.0 environments.
- Contributed to the "Swarm Robotics for Industry 4.0" project supported by the Independent Research Fund Denmark under grant 0136-00251B.

Robotics Engineer (Freelance) - AiGro Netherlands (2021-2022)
Remote
Summary: Designed simulations and developed localization algorithms for agricultural mobile robots.
Highlights:

- Built and tested a 2D simulator designed to help with developing pathfinding and localization algorithms for autonomous mobile robots.
- Implemented a sensor fusion approach based on Kalman filter to localize an autonomous mobile robot.
- Used the robot's IMU and GPS data collected in real-life experiments to test and validate the localization approach.

Temporary Lecturer - University of 20 Aout 1955 (2019-2022)
Skikda, Algeria
Summary: I earned extra credit for teaching laboratory sessions and tutorial classes as part of my PhD.
Highlights:

- Taught advanced signal processing and digital regulation both as laboratory sessions and tutorial classes.
- Supervised master's students' final-year projects.

Education

Ph.D - Automation (2019-2023)
University of Skikda, Algeria
Thesis title: Contributions to the command of mobile robots.
Note: Worked mainly on mobile robots pathfinding.

Master's degree - Instrumentation (2017-2019)
University of Skikda, Algeria
Related coursework: Sensors, advanced digital electronics, digital regulation, microprocessors and DSP, electronic circuits.

Bachelor degree - Electronics Engineering (2014-2017)
University of Skikda, Algeria
Related coursework: fundamental electronics, micro-processor systems, signal processing, OOP.

Research Publications

Journal Articles

1. M. Boumediene, A. Maoudj, and A. L. Christensen, "HM-DRL: Enhancing multi-agent pathfinding with a heatmap-based heuristic for distributed deep reinforcement learning," *Applied Intelligence*, vol. 55, p. 873, Jul. 2025. DOI: 10.1007/s10489-025-06747-0.

2. M. Boumediene, L. Mehennaoui, and A. Lachouri, "FDA*: A Focused Single-Query Grid Based Path Planning Algorithm," *Journal of Automation, Mobile Robotics and Intelligent Systems (JAMRIS)*, pp. 37-43, May 2022. DOI: 10.14313/JAMRIS/3-2021/17.

3. M. Boumediene, A. L. Christensen, and A. Maoudj, MAPF-Relocate: Mitigating Congestion through Strategic Agent Relocation in Multi-Agent Pathfinding, Status: revision invited, Springer's *Journal of Intelligent and Robotic Systems*.

Conference Proceedings

1. I. Panov, M. Boumediene, H. Midtiby, and K. Jensen, "Gen4jectory algorithm - 4-D trajectory planning with minimised flight time for multiple rotary-wing UAVs," 15th Annual International Micro Air Vehicle Conference and Competition (IMAV 2024-25), Sep. 2024, pp. 242-252. URL: `https://www.imavs.org/imav2024-proceedings/`.

2. C. Bellel, L. Mehennaoui, F. Bouchareb, and M. Boumediene, "Unicycle mobile robot control using neural model predictive controller," Fourth International Conference on Technological Advances in Electrical Engineering (ICTAEE'23), Laboratory of Automatic of Skikda, 20 Aout 1955 University, Skikda, Algeria, May 2023, pp. 110-117. URL: `https://drive.google.com/file/d/1C6RUU5zixy_UfJga0yu86lvQMPaJo1EE/view?usp=sharing`.

3. M. Boumediene, S. Laouar, D. L. Mehennaoui, and P. S. Ouchtati, "Design, simulation and control of a self-balancing robot in a Gazebo environment and ROS 2 framework," Proceedings of ICTAEE, Skikda, Algeria, May 2023, pp. 98-103. URL: `https://drive.google.com/file/d/1C6RUU5zixy_UfJga0yu86lvQMPaJo1EE/view?usp=sharing`.

4. M. Boumediene, N. Zeghida, B. Manaa, and D. L. Mehennaoui, "Design, construction, and control of a self-balancing robot including a new frame assembly approach and a custom PCB," Proceedings of ICTAEE, Skikda, Algeria, May 2023, pp. 200-206. URL: `https://drive.google.com/file/d/1C6RUU5zixy_UfJga0yu86lvQMPaJo1EE/view?usp=sharing`.

Published Courses

1. M. Boumediene, “Drone Simulation Using ROS 2 and Gazebo,” Udemy, Nov. 2025. URL: `https://www.udemy.com/course/ros2-drone-simulation-with-gazebo/`

References

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